PROJECT REPORT

Course: Fundamentals of AI & ML (CSA2001)

Project Title: Campus Placement Predictor using Logistic Regression

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Slot: B11 + B12 + B13

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1. Introduction & Problem Statement

In an elite technical institution like **VIT Bhopal**, campus placements are the most significant milestone for engineering students. However, this period is often accompanied by high levels of anxiety and uncertainty. Students frequently struggle to evaluate if their current academic standing (CGPA) and extracurricular achievements (Internships/Projects) are sufficient to meet the criteria of "Dream" or "Super Dream" companies.

The Problem: There is a lack of accessible, data-driven tools that allow students to simulate their placement probability based on historical trends. Students often rely on hearsay rather than statistical evidence.

Objective: To build a Machine Learning-based classification tool that predicts placement probability with a confidence score, helping students identify areas of improvement before the actual recruitment season begins.

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2. Why This Matters

This project serves three critical purposes:

1. **Anxiety Reduction:** By providing a "Probability Score," students get a realistic view of their standing.
2. **Targeted Improvement:** The tool helps students see how much an additional internship or a 0.5 increase in CGPA could change their outcome.
3. **Application of AI:** It demonstrates how a fundamental ML algorithm like **Logistic Regression** can be applied to solve a local, real-world community problem.

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3. My Approach (The ML Pipeline)

I followed a structured Machine Learning workflow to ensure the solution was robust and scalable:

A. Data Generation

Since real-time placement data is confidential, I created a synthetic dataset (placement_data.csv) of 1000 student records. I designed the logic to mirror VIT's general trends, where weightage is distributed across CGPA (40%), Internships (30%), and Technical Skills (30%).

B. Feature Engineering

I selected five key features that impact placement:

- **CGPA:** Academic consistency.
- **Internships:** Practical industry exposure.
- **Projects:** Application of theoretical knowledge.
- **Technical Skills Score:** Performance in coding rounds.
- **Soft Skills Score:** Performance in HR/Interview rounds.

C. Algorithm Selection

I chose **Logistic Regression** because the problem is a **Binary Classification** task (the output is either "Placed" or "Not Placed"). Logistic Regression is ideal here as it provides not just a label, but a probability score (via the Sigmoid function).

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4. Key Decisions & Implementation

- **Model Persistence:** I decided to use the joblib library to "serialize" the model. This means the model is trained once and saved as placement_model.pkl. This allows the application to load instantly without retraining, making it feel like a real software product.
- **User Interface:** I implemented a clean command-line interface (CLI) that guides the user through the input process, ensuring that even a non-technical user can interact with the AI.
- **Data Validation:** I used **Pandas DataFrames** for user input to ensure the model receives feature names exactly as it did during training, preventing errors.

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5. Challenges Faced

1. **The "Feature Names" Warning:** Initially, the model threw UserWarnings because I was passing raw arrays for prediction. I learned that Scikit-Learn models prefer consistency between training and testing data structures. I solved this by wrapping user input in a named DataFrame.
2. **Balancing the Dataset:** Ensuring the synthetic data wasn't too biased (e.g., making sure a high CGPA student could still "fail" if their skills were zero) required fine-tuning the data generation logic.

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6. What I Learned (Reflection)

This capstone project was a transformative experience. Key takeaways include:

- **End-to-End Workflow:** AI is more than just an algorithm; it involves data cleaning, model training, saving, and building a user interface.
- **Mathematical Logic:** I now deeply understand how the Sigmoid function maps linear equations to a range of 0 to 1 for classification.
- **Documentation:** I realized that a great project is only as good as its README and Report. Clear communication is as important as clean code.

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7. Conclusion

The **Campus Placement Predictor** successfully achieves its goal of providing a reliable, data-backed prediction for students. While the current version uses synthetic data, the architecture is ready to be trained on actual VIT placement records in the future.

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8. Appendix (Screenshots)

Code:

main.py > ...

```python
import pandas as pd
import numpy as np
import os
import joblib
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
print("Welcome! \n")
print("Hello! I am SARTHAK JINDAL (25BCE10189) presenting you my ML project....\n")


DATA_FILE = 'placement_data.csv'
MODEL_FILE = 'placement_model.pkl'
FEATURE_NAMES = ['CGPA', 'Internships', 'Projects', 'Technical_Skills_Score', 'Soft_Skills_Score']

def train_model():
    print("\n--- Initializing Machine Learning Model ---")
    if not os.path.exists(DATA_FILE):
        print(f"Error: {DATA_FILE} not found. Please ensure your dataset is in the same folder.")
        return None


    df = pd.read_csv(DATA_FILE)
    X = df[FEATURE_NAMES]
    y = df['Placed']


    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)


    model = LogisticRegression()
    model.fit(X_train, y_train)


    predictions = model.predict(X_test)
    acc = accuracy_score(y_test, predictions)
    print(f"Status: Model trained successfully.")
    print(f"Model Accuracy: {acc*100:.2f}%")


    joblib.dump(model, MODEL_FILE)
    return model

def predict_placement(model):
    print("\n" + "="*40)
    print("      VIT PLACEMENT PREDICTOR 2026      ")
    print("="*40)

    try:

        cgpa = float(input("Enter Student CGPA (e.g., 8.5): "))
        interns = int(input("Number of Internships completed: "))
        projects = int(input("Number of Projects completed: "))
        tech_score = int(input("Technical Skill Score (1-100): "))
        soft_score = int(input("Soft Skill Score (1-100): "))


        user_input_df = pd.DataFrame([[cgpa, interns, projects, tech_score, soft_score]],
                                     columns=FEATURE_NAMES)


        prediction = model.predict(user_input_df)
        probability = model.predict_proba(user_input_df)[0][1]

        print("\n--- PREDICTION ANALYSIS ---")
        if prediction[0] == 1:
            print(f"STATUS: PLACED!")
            print(f"PROBABILITY: {probability*100:.1f}%")
            print("ADVICE: Excellent profile. You are eligible for Super Dream offers.")
        else:
            print(f"STATUS: NOT PLACED YET")
            print(f"PROBABILITY: {(1-probability)*100:.1f}% (Chance of rejection)")
            print("ADVICE: Consider improving CGPA or adding one more internship.")

    except ValueError:
        print("\n[!] ERROR: Please enter valid numeric values.")

if __name__ == "__main__":
```

```python
if __name__ == "__main__":

    if os.path.exists(MODEL_FILE):
        placement_model = joblib.load(MODEL_FILE)
    else:

        placement_model = train_model()

    if placement_model:

        while True:
            predict_placement(placement_model)
            print("-" * 40)
            cont = input("Check another student? (y/n): ").lower()
            if cont != 'y':
                print("\nThank you for using the Predictor. Good luck !")
                break
```

OUTPUT:

```
Welcome!

Hello! I am SARTHAK JINDAL (25BCE10189) presenting you my ML project....


========================================
      VIT PLACEMENT PREDICTOR 2026
========================================
Enter Student CGPA (e.g., 8.5): 9.1
Number of Internships completed: 4
Number of Projects completed: 6
Technical Skill Score (1-100): 80
Soft Skill Score (1-100): 75

--- PREDICTION ANALYSIS ---
STATUS: PLACED!
PROBABILITY: 100.0%
ADVICE: Excellent profile. You are eligible for Super Dream offers.
----------------------------------------
Check another student? (y/n): y

========================================
      VIT PLACEMENT PREDICTOR 2026
========================================
Enter Student CGPA (e.g., 8.5): 6.4
Number of Internships completed: 3
Number of Projects completed: 4
Technical Skill Score (1-100): 50
Soft Skill Score (1-100): 40

--- PREDICTION ANALYSIS ---
STATUS: PLACED!
PROBABILITY: 99.8%
ADVICE: Excellent profile. You are eligible for Super Dream offers.
----------------------------------------
Check another student? (y/n): y
```

```
========================================
      VIT PLACEMENT PREDICTOR 2026
========================================
Enter Student CGPA (e.g., 8.5): 3
Number of Internships completed: 0
Number of Projects completed: 1
Technical Skill Score (1-100): 20
Soft Skill Score (1-100): 10

--- PREDICTION ANALYSIS ---
STATUS: NOT PLACED YET
PROBABILITY: 100.0% (Chance of rejection)
ADVICE: Consider improving CGPA or adding one more internship.
----------------------------------------
Check another student? (y/n): n

Thank you for using the Predictor. Good luck !
```